DAMIEN CHEMISTRY OLYMPIAD NOTES

Acid–Base Theories

Conceptual Question Set

Recommended balance: 80% conceptual : 20% calculation
Coverage: Arrhenius, Brønsted–Lowry and Lewis theories; conjugate pairs; amphiprotic species; non-aqueous acid–base chemistry; strength versus concentration; basicity versus al$K_a$linity; boric acid; solvent levelling.

Foundation for SJChO and advanced H2 Chemistry

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1. Arrhenius Acids and Bases

Concept focus. Acids increase aqueous H_3O^+ ; bases increase aqueous OH^- . Always state whether the definition is being used strictly or operationally.

Question 1 Easy

Classify HCl(aq) and NaOH(aq) under the Arrhenius theory. Write an equation showing how each increases the concentration of the characteristic ion in water.

Question 2 Medium

A sample containing 0.0150 mol of $Ca(OH)_2$ is dissolved to make 500 cm^3 of solution. Assuming complete dissociation, calculate $[OH^-]$ and explain why $Ca(OH)_2$ is an Arrhenius base.

Question 3 Hard

Sodium oxide reacts with water according to $Na_2O + H_2O \rightarrow 2\ NaOH$. Is solid Na_2O itself an Arrhenius base? Give a careful answer rather than a one-word classification.

2. Limitations of the Arrhenius Model

Concept focus. Test the definition against non-aqueous reactions, bases without hydroxide groups and electron-pair reactions without proton transfer.

Question 4 Easy

Explain why the reaction $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$ cannot be adequately described using only the Arrhenius theory.

Question 5 Medium

The reaction $BF_3 + NH_3 \rightarrow BF_3 \cdot NH_3$ contains neither transferable protons nor hydroxide ions. Why is it nevertheless an acid–base reaction, and what limitation of Arrhenius theory does it demonstrate?

Question 6 Hard

Critique the statement: “An acid is any substance containing hydrogen, while a base is any substance containing an OH group.” Give at least two chemically distinct counterexamples.

3. Brønsted–Lowry Proton Transfer

Concept focus. An acid donates H^+ ; a base accepts H^+ . Track one proton from reactant to product.

Question 7 Easy

In $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, identify the Brønsted–Lowry acid and base. State what is transferred.

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Question 8 Medium

For $HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$, determine whether water acts as an acid or a base, and justify your choice by tracking the proton.

Question 9 Hard

Consider $H_2PO_4^- + NH_3 \rightleftharpoons HPO_4^{2-} + NH_4^+$. The relevant values are pK_a $(H_2PO_4^-) = 7.20$ and $pKa(NH_4^+) = 9.25$. Identify the acid and base on each side and estimate the equilibrium constant for the forward reaction.

4. Conjugate Acid—Base Pairs

Concept focus. A conjugate pair differs by exactly one H^+ ; charge changes by one unit.

Question 10 Easy

Give the conjugate acid of NH_3 and the conjugate base of H_2O.

Question 11 Medium

In $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$, identify both conjugate acid—base pairs.

Question 12 Hard

In water, the hexaaquairon(III) ion undergoes $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$. Identify the conjugate pairs and explain why a metal aquo ion can behave as a Brønsted acid even though the proton originally belongs to a coordinated water molecule.

5. Amphiprotic Species

Concept focus. An amphiprotic particle has both a proton it can donate and a site that can accept a proton.

Question 13 Easy

Why is HCO_3^- amphiprotic? Name the species formed when it accepts a proton and when it donates a proton.

Question 14 Medium

Write two equations with water that demonstrate the amphiprotic behaviour of $H_2PO_4^-$.

Question 15 Hard

For HS^-, K_a for $HS^- \rightarrow H^+ + S^{2-}$ is approximately 10^{-14}, while K_a for H_2S is approximately 10^{-7}. Use $Kw = 10^{-14}$ to decide whether HS^- shows the stronger tendency to act as an acid or as a base in water.

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6. Relative Acid and Base Strength

Concept focus. Proton transfer favours formation of the weaker acid and weaker base; larger K_a corresponds to stronger acid.

Question 16 Easy

Hydrochloric acid is essentially fully ionised in dilute water, whereas ethanoic acid is only partially ionised. Which is the stronger acid, and what does "stronger" mean here?

Question 17 Medium

For $CH_3COOH + F^- \rightleftharpoons CH_3COO^- + HF$, use pK_a (CH_3COOH) = 4.76 and pK_a (HF) = 3.17 to estimate K and state which side is favoured.

Question 18 Hard

Arrange $HClO$, $HClO_2$, $HClO_3$ and $HClO_4$ in increasing acid strength. Explain the trend without merely quoting oxidation states.

7. Lewis Acids as Electron-Pair Acceptors

Question 19 Easy

In $BF_3 + NH_3 \rightarrow BF_3 \cdot NH_3$, identify the Lewis acid and state why it can accept an electron pair.

Question 20 Medium

Compare the Lewis acidity of $AlCl_3$ and $AlCl_4^-$. Which is the better electron-pair acceptor, and why?

Question 21 Hard

A simplistic argument predicts BF_3 to be more Lewis acidic than BCl_3 because fluorine is more electronegative. Yet toward many common donors, BCl_3 is the stronger Lewis acid. Explain this apparent reversal.

8. Lewis Bases as Electron-Pair Donors

Question 22 Easy

Which of NH_3, H_2O and Cl^- can act as Lewis bases? Identify the donating electron pair in each case.

Question 23 Medium

Predict whether NH_3 or NF_3 is the stronger Lewis base toward BF_3. Explain using electron availability.

Question 24 Hard

Pyridine and pyrrole both contain a nitrogen lone pair. In pyridine the lone pair lies outside the aromatic π system; in pyrrole it supplies two electrons to complete the six-electron aromatic π system. Which is the stronger Lewis base, and why?

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9. Coordinate-Bond Formation

Concept focus. A donor supplies both electrons to an acceptor; after formation, the bond is not a separate physical category of covalent bond.

Question 25 Easy

Show the formation of NH_4^+ from NH_3 and H^+ using a coordinate-bond arrow. Which atom supplies both bonding electrons?

Question 26 Medium

Gaseous $AlCl_3$ can dimerise to Al_2Cl_6. Explain how chloride bridges form using the language of coordinate bonding.

Question 27 Hard

In $[Cu(NH_3)_4(H_2O)_2]^{2+}$, identify the Lewis acid, Lewis bases and the number of coordinate bonds. After formation, are these bonds fundamentally different from ordinary covalent bonds?

10. Species Satisfying One Definition but Not Another

Question 28 Easy

Classify BF_3 under the Arrhenius, Brønsted–Lowry and Lewis definitions of an acid.

Question 29 Medium

Classify $NH_3(g)$ as a base under the three theories. Then explain why its classification may appear to change when it is dissolved in water.

Question 30 Hard

Classify $CO_2 + OH^- \rightarrow HCO_3^-$ under the three theories. Why is calling CO_2 an "acidic oxide" compatible with its having no proton?

11. Acid–Base Behaviour Outside Aqueous Solution

Question 31 Easy

Explain the acid–base roles in $HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$ without referring to H_3O^+ or OH^- .

Question 32 Medium

Liquid ammonia undergoes autoionisation: $2\ NH_3 \rightleftharpoons NH_4^+ + NH_2^-$. In this solvent, explain why NH_4Cl behaves as an acid and KNH_2 behaves as a base.

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Question 33 Hard

In glacial ethanoic acid, a very strong acid can protonate the solvent: $HClO_4 + CH_3COOH \rightarrow ClO_4^- + CH_3COOH_2^+$. Identify the acid and base, and explain why an aqueous-only definition is inadequate.

12. Acid Strength versus Acid Concentration

Question 34 Easy

Two solutions contain 0.100 mol dm^{-3} and 1.00 mol dm^{-3} HCl, respectively. Do they contain acids of different strength? Explain.

Question 35 Medium

A 1.0 mol dm^{-3} solution of ethanoic acid and a 0.010 mol dm^{-3} solution of hydrochloric acid are compared. State separately which acid is stronger and which solution contains more acid particles per unit volume.

Question 36 Hard

A weak monoprotic acid HA has $K_a = 1.0 \times 10^{-5}$ and concentration 0.100 mol dm^{-3}.
Compare its equilibrium $[H_3O^+]$ with that of a 5.0×10^{-4} mol dm^{-3} strong monoprotic acid. Use the weak-acid approximation but do not calculate pH.

13. Basicity versus Alkalinity

Question 37 Easy

Explain why CuO is a base but not an alkali, whereas NaOH is both.

Question 38 Medium

One litre of 0.0100 mol dm^{-3} NaOH and one litre of 0.200 mol dm^{-3} NH_3 are separately neutralised by a strong acid. Which contains the intrinsically stronger base particle, and which can neutralise more moles of H^+ ?

Question 39 Hard

A 250 cm3 solution contains 0.0200 mol dm^{-3} CO_3^{2-} and 0.0300 mol dm^{-3} HCO_3^- . Calculate the maximum moles of H^+ it can consume when both species are converted fully to H_2CO_3. Relate the result to alkalinity.

14. Olympiad Extension: Boric Acid as a “Weird” Acid

Question 40 Easy

Boric acid reacts with water as follows: $B(OH)_3 + 2\ H_2O \rightleftharpoons B(OH)_4^- + H_3O^+$. Explain why $B(OH)_3$ is acting primarily as a Lewis acid rather than by directly donating one of its own O—H protons.

Question 41 Medium

A sample contains 0.0100 mol of $B(OH)_3$. How many moles of NaOH are required for complete neutralisation according to its aqueous mechanism? Explain why the answer is not three times larger.

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Question 42 Hard

Boric acid is placed in water enriched with $H_2{}^{18}O$. Using the Lewis-acid mechanism, predict whether an ^{18}O label can become incorporated into $B(OH)_4^-$. Explain what this mechanistic test distinguishes from a simple direct-proton-donation picture.

15. Olympiad Extension: Solvent Levelling and Differentiation

Question 43 Easy

Why do HCl, HBr, HI and $HClO_4$ all appear essentially completely ionised in dilute aqueous solution, even though their intrinsic gas-phase acidities are not identical?

Question 44 Medium

Liquid ammonia autoionises to NH_4^+ and NH_2^- . Identify the strongest acid and strongest base that can persist in this solvent, and explain what happens when a much stronger acid or base is added.

Question 45 Hard

Water cannot readily distinguish the strengths of $HClO_4$ and HNO_3, but a less basic solvent such as glacial ethanoic acid may do so. Explain what makes a solvent “differentiating” and why changing solvent can reverse or reveal apparent strength orderings.

End of question set

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